

Niranjan Rao

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PROFESSIONAL SUMMARY

2.5+ years of experience building and deploying large-scale AI systems in marketing and retail. Skilled in developing embeddings, optimization models, and generative AI tools using PyTorch, PySpark, and Vertex AI. Strong focus on measurable business impact and scalable ML architectures.

EDUCATION

Indian Institute of Science

Masters in Data Science and Business Analytics (CGPA: 8.7)

Bengaluru, KA

Aug. 2021 – June 2023

Birla Institute of Technology and Science, Pilani

B.E., Electrical & Electronics Engg. (CGPA: 8.4)

Pilani, RJ

Aug. 2016 – May 2020

EXPERIENCE

Senior Data Scientist - Offer Personalization

Apr 2024 – present

Target

Bengaluru, KA

- Developed **high-dimensional entity embeddings** for recommendation and coupon-prediction systems, enabling **personalized guest experiences** across digital and in-store platforms. Integrated the model into a **production-scale Vertex AI pipeline** for continuous inference. Constructed a **Kubernetes Flow Pipeline (KFP)** in **Google Vertex AI**, using **Dataproc** for distributed Spark processing, and **DSL serverless** for model training.
- Researched and implemented graph neural network (GNN) model architectures - **MetaPath2Vec**, **Graph Autoencoders**, **LightGCN**, and **Graph Transformers**, achieving **75%** recall and **78%** avg. precision in link prediction tasks.

Senior Data Scientist - Supply Chain

Aug 2023 - Mar 2024

Target

Bengaluru, KA

- Designed and implemented an optimization engine to perform route selection for import purchase orders, by formulating a **mixed-integer linear programming problem** using the **CBC** solver within the **PuLP** package, optimizing for transportation cost.
- Compared and contrasted the route selection and overall transportation cost after imposing an extra penalty term for early arrival.
- The optimized route selection resulted in the reduction of transportation costs by **11%** (**9%** with penalization) compared to a heuristic route selection algorithm currently in use.

Quantitative Research Intern

Dec 2022 – June 2023

State Street Global Advisors

Bengaluru, KA

- Identified and analyzed key factors impacting fixed-income portfolios like credit spread. Used PCA and regression analysis to isolate significant risk factors and their effect on portfolio performance.

TECHNICAL SKILLS

Languages: C/C++, Python, R, SQL

ML/AI Frameworks: PyTorch, PyTorch-Geometric, scikit-learn, TensorFlow

LLM/GenAI Frameworks: LangChain, LlamaIndex, ChromaDB, GPTs, Google ADK, Vertex AI LLM APIs

Big Data & MLOps: PySpark, Apache Spark, Kubernetes Flow Pipelines, Dataproc, Docker

Cloud & Tools: Google Cloud Platform, Vertex AI, Git, JupyterLab, VS Code, GitHub Copilot, Streamlit

SELECTED AI PROJECTS (MORE ON [WEBSITE](#))

Multi-Agent AI System for Perishable Inventory Simulation | *Python, GoogleADK, GPT-5.1, Streamlit*

May 2026

- Built a multi-agent AI chatbot using **Google Agent Development Kit (ADK)**, **OpenAI API (GPT-5.1)**, and **HDFS** to automate perishables inventory simulations for store order inflow/outflow scenarios.
- Engineered an end-to-end orchestration pipeline with **prompt parsing**, **tool calling**, **memory**, and **asynchronous agent handoffs** to extract simulation parameters, execute simulation runs, and manage execution autonomously.
- Generated queries to user prompts by using analysis agent to deliver insights grounded on simulation runs. Observed **33%** reduction in queries by business teams due to the self-serve nature of the agentic system.

Uplift Modelling | *Python, causalm*

Jul 2024

- Strategized a novel method to isolate persuadable purchasers using causal learning methodologies to estimate **CATE**.
- Fitted an **XGBT Meta-Learner** model using the A/B test results of a marketing campaign targetting **3.6M** treatment and **270K** control guests.
- Tested out results on a "lookalike" campaign, and obtained **70%** of the predicted uplift by reaching out to **30%** of the guests, resulting in savings of **\$700K** in discounted markdowns.